WORKED SOLUTION

STEP III Specimen paper, Question 14

Mechanics • Rotational Collisions

STEP III Specimen paper, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q14

The key idea

The pivot exerts an unknown impulse on the rod, so linear momentum is *not* conserved, but that impulse has no moment about the pivot: angular momentum about the midpoint is conserved. Together with Newton's law of restitution (coefficient 1, since the collision is perfectly elastic) applied at the point of contact, that fixes everything. For the second part, find what the particle does next and follow both bodies: the far half of the rod comes round to the particle's line.

Solution

Model. The particle is a point mass; the table is smooth, so there's no friction on either body; the rod is uniform and pivoted smoothly at its midpoint O. The moment of inertia of the rod about O is

$$I = \tfrac{1}{12}m(4a)^2 = \tfrac{4}{3}ma^2.$$

During the collision the impulses act along the particle's line of motion (the rod is smooth and the particle hits it at right angles), so the particle keeps moving along the same line. Let its velocity after the collision be v, measured in its original direction of motion.

The angular speed

Angular momentum about O. The only external impulse on the system (particle and rod) comes from the pivot at O, and it has zero moment about O. Before the collision the particle's angular momentum about O is mua and the rod is at rest, so

$$mua = mva + \tfrac{4}{3}ma^2\omega \qquad \Longrightarrow \qquad u = v + \tfrac{4}{3}a\omega. \tag{1}$$

Restitution. The point of the rod that is struck is at distance a from O, so just after the collision it moves at speed $a\omega$ in the particle's original direction. With coefficient of

restitution 1, the speed of separation equals the speed of approach:

$$a\omega - v = u. \tag{2}$$

Ask yourself Can we use Newton's law of restitution for a rotating body?

Yes: it applies to the velocities of the two points in contact, along the common normal. If you prefer, use energy instead: "perfectly elastic" means kinetic energy is conserved, $\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + \frac{1}{2}\cdot\frac{4}{3}ma^2\omega^2$. Writing this as $m(u-v)(u+v) = \frac{4}{3}ma^2\omega^2$ and using (1), $u - v = \frac{4}{3}a\omega$, we get $u + v = a\omega$ (dividing by $\omega \neq 0$), which is exactly (2).

Adding (1) and (2) eliminates v:

$$2u = \tfrac{7}{3}a\omega \qquad \Longrightarrow \qquad a\omega = \frac{6u}{7},$$

as required. Then from (2),

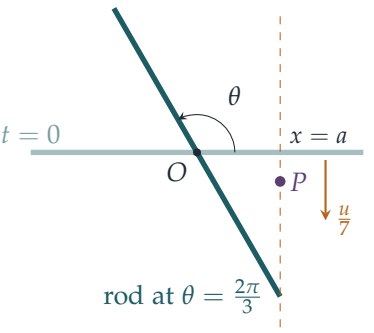
$$v = a\omega - u = -\frac{u}{7},$$

so the particle bounces back along its original line with speed $u/7$.

The second collision

Take axes on the table with origin O, the x-axis along the rod's initial position and the particle originally moving in the positive y-direction along the line $x = a$, so that it strikes the rod at $(a, 0)$ at time $t = 0$. Then for $t > 0$:

- the particle is at $\left(a,\ -\frac{u}{7}t\right)$, moving steadily in the negative y-direction;
- the struck point moved in the positive y-direction, so the rod turns anticlockwise, and at time t it makes an angle $\theta = \omega t$ with the x-axis. No further force acts on either body until they collide again, so both motions are uniform.



The rod has half-length $2a$, so it reaches the line $x = a$ only when the point of the rod on that line, at distance $a/|\cos\theta|$ from O, is within $2a$ of O; that is, when $|\cos\theta| \geq \frac{1}{2}$. Where it does, the rod crosses the line at height $y = a\tan\theta$.

- For $0 < \theta \le \frac{\pi}{3}$ the rod meets the line at $y = a\tan\theta > 0$, above the x-axis, while the particle is below it: no contact.
- For $\frac{\pi}{3} < \theta < \frac{2\pi}{3}$ the rod doesn't reach the line at all.
- For $\frac{2\pi}{3} \le \theta < \pi$ the *other* half of the rod crosses the line, at $y = a\tan\theta < 0$, on the same side as the particle.

Now compare positions. Since $t = \theta/\omega = \frac{7a\theta}{6u}$, the particle's y-coordinate at angle θ is $-\frac{u}{7}t = -\frac{a\theta}{6}$. Consider

$$f(\theta) = (\text{rod's } y \text{ on } x = a) - (\text{particle's } y) = a\tan\theta + \frac{a\theta}{6}, \qquad \tfrac{2\pi}{3} \le \theta < \pi.$$

- At $\theta = \frac{2\pi}{3}$ (the end of the rod reaches the line): $f = a\left(-\sqrt{3} + \frac{\pi}{9}\right) < 0$, since $\frac{\pi}{9} < 1 < \sqrt{3}$. The particle is at $y = -\frac{\pi}{9}a \approx -0.35a$, between the x-axis and the end of the rod at $y = -\sqrt{3}a$: the rod's end has come round *beyond* the particle.
- As $\theta \to \pi^-$: $\tan\theta \to 0$, so $f \to \frac{\pi}{6}a > 0$. The rod is sweeping back up to the x-axis, while the particle is at about $-0.52a$.

f is continuous on $[\frac{2\pi}{3}, \pi)$, so by the intermediate value theorem $f(\theta) = 0$ for some θ in between: at that instant the rod is at the particle's position, and the two collide again. (No earlier contact is possible, by the first two bullet points above.) This completes the proof.

Numerically, the collision happens where $\tan\theta = -\theta/6$, at $\theta \approx 2.716$ (about $155.6°$), at time $t \approx 3.17\,a/u$, with the particle at $y \approx -0.45a$, about $1.10a$ from O along the rod.

Checks

- Energy: with $a\omega = \frac{6}{7}u$ and $v = -\frac{1}{7}u$, the final kinetic energy is $\frac{1}{2}m\left(\frac{1}{49} + \frac{4}{3}\cdot\frac{36}{49}\right)u^2 = \frac{1}{2}m\cdot\frac{49}{49}u^2$, exactly the initial energy.
- Repeating the argument for a strike at distance d on a body of moment of inertia I gives $d\omega = 2u/(1 + I/md^2)$. As $I \to 0$ the struck point moves off at $2u$, as for an elastic collision with a very light body; as $I \to \infty$ the rod hardly turns and the particle bounces back with speed u. Here $I/ma^2 = \frac{4}{3}$ gives $\frac{6}{7}u$.
- We checked with exact fractions that $a\omega = \frac{6}{7}u$, $v = -\frac{1}{7}u$ conserve both angular momentum and energy, then stepped both motions forward in time: the first contact after the collision occurs at $\theta \approx 2.7165$, agreeing with a bisection solution of $\tan\theta + \theta/6 = 0$.

Common slips

- Conserving *linear* momentum. The pivot supplies an impulse, so only angular momentum about the pivot is conserved.

- Using $I = \frac{1}{3}m\ell^2$ (the value about an end) or forgetting that the rod's length is $4a$, not $2a$.
- Looking for the second collision on the half of the rod that was struck. That half moves away from the particle's line; it's the other half, coming round, that meets it.

Beyond STEP

Pairing angular momentum about a fixed point with an energy or restitution condition is the standard way to handle impulsive motion of rigid bodies, and leads on to the idea of the *centre of percussion*: the point at which a pivoted body can be struck with no impulse at the pivot. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body collisions and angular momentum carefully and has many problems of exactly this kind.

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